

Complex Numbers



Arithmetic with complex numbers

- 1 Let $z_1 = 4 - 2i$ and $z_2 = -3 + i$. Find $z_1 + z_2$ and $z_1 z_2$.
 - 2 Find $(6 + i) - (2 - 5i)$.
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Complex roots of a quadratic

- 3 Solve $x^2 + 6x + 13 = 0$.
 - 4 Solve $2x^2 - 4x + 10 = 0$.
 - 5 Without solving it, explain how you know $x^2 + 2x + 5 = 0$ has complex (not real) roots.
 - 6 Verify that $x = -3 + 2i$ is a root of $x^2 + 6x + 13 = 0$ by direct substitution.
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Modulus

- 7 Find $|z|$ for $z = -3 + 2i$ (one of the roots found above).
 - 8 Find $|z|$ for $z = 5 - 12i$.
 - 9 Explain why a conjugate pair $a \pm bi$ always has the same modulus for both roots.
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Multiple choice — easy

- 10 Let $z_1 = -6 + 9i$ and $z_2 = 2 + 6i$. Find $z_1 + z_2$.
 - a. $-2 + 15i$
 - b. $-4 + 13i$
 - c. $-4 + 15i$
 - d. $-8 + 3i$

11 Let $z_1 = -2 - 3i$ and $z_2 = -4 - 8i$. Find $z_1 + z_2$.

a. $-6 - 13i$

b. $-6 - 11i$

c. $2 + 5i$

d. $-4 - 11i$

12 Let $z_1 = 3 + 8i$ and $z_2 = -9 - 7i$. Find $z_1 + z_2$.

a. $12 + 15i$

b. $-6 - i$

c. $-4 + i$

d. $-6 + i$

13 Let $z_1 = 7 + 9i$ and $z_2 = -5 + 5i$. Find $z_1 + z_2$.

a. $2 + 12i$

b. $2 + 14i$

c. $12 + 4i$

d. $4 + 14i$

14 Let $z_1 = -8 - 8i$ and $z_2 = 5 + 3i$. Find $z_1 + z_2$.

a. $-13 - 11i$

b. $-3 - 5i$

c. $-1 - 5i$

d. $-3 - 7i$

- 15 Let $z_1 = 8 + 5i$ and $z_2 = 9 - 3i$. Find $z_1 + z_2$.
- a. $19 + 2i$
 - b. $17 + 2i$
 - c. 17
 - d. $-1 + 8i$
- 16 Let $z_1 = -7 - 3i$ and $z_2 = -4 + 2i$. Find $z_1 + z_2$.
- a. $-3 - 5i$
 - b. $-11 - 3i$
 - c. $-9 - i$
 - d. $-11 - i$
- 17 Let $z_1 = 3 - 7i$ and $z_2 = -1 - i$. Find $z_1 + z_2$.
- a. $2 - 8i$
 - b. $4 - 6i$
 - c. $4 - 8i$
 - d. $2 - 10i$
- 18 Let $z_1 = 1 + 8i$ and $z_2 = -2 - 4i$. Find $z_1 + z_2$.
- a. $3 + 12i$
 - b. $1 + 4i$
 - c. $-1 + 2i$
 - d. $-1 + 4i$

19 Let $z_1 = 5 - i$ and $z_2 = -3 - 6i$. Find $z_1 + z_2$.

a. $2 - 9i$

b. $8 + 5i$

c. $2 - 7i$

d. $4 - 7i$

20 Let $z_1 = 3 + 7i$ and $z_2 = -9 + 4i$. Find $z_1 + z_2$.

a. $-4 + 11i$

b. $-6 + 9i$

c. $-6 + 11i$

d. $12 + 3i$

21 Let $z_1 = 5 + 2i$ and $z_2 = 8 - 2i$. Find $z_1 + z_2$.

a. 13

b. $-3 + 4i$

c. $13 - 2i$

d. 15

22 Let $z_1 = 9 - 7i$ and $z_2 = -4 - 8i$. Find $z_1 + z_2$.

a. $7 - 15i$

b. $5 - 15i$

c. $13 + i$

d. $5 - 17i$

23 Let $z_1 = -8 - i$ and $z_2 = -5 + 6i$. Find $z_1 + z_2$.

a. $-13 + 3i$

b. $-3 - 7i$

c. $-11 + 5i$

d. $-13 + 5i$

24 Let $z_1 = -3 + 5i$ and $z_2 = 8 + 3i$. Find $z_1 + z_2$.

a. $7 + 8i$

b. $5 + 6i$

c. $-11 + 2i$

d. $5 + 8i$

Multiple choice — medium

25 Solve $x^2 - 2x + 5 = 0$.

a. $1 \pm 2i$

b. $1 \pm 3i$

c. $2 \pm 2i$

d. $-1 \pm 2i$

26 Solve $x^2 - 10x + 74 = 0$.

a. $5 \pm 8i$

b. $6 \pm 7i$

c. $-5 \pm 7i$

d. $5 \pm 7i$

27 Solve $x^2 + 2x + 65 = 0$.

a. $-1 \pm 9i$

b. $-1 \pm 8i$

c. $1 \pm 8i$

d. $0 \pm 8i$

28 Solve $x^2 + 12x + 72 = 0$.

a. $6 \pm 6i$

b. $-6 \pm 6i$

c. $-6 \pm 7i$

d. $-5 \pm 6i$

29 Solve $x^2 + 8x + 32 = 0$.

a. $-4 \pm 4i$

b. $-4 \pm 5i$

c. $-3 \pm 4i$

d. $4 \pm 4i$

30 Solve $x^2 - 6x + 58 = 0$.

a. $3 \pm 7i$

b. $4 \pm 7i$

c. $-3 \pm 7i$

d. $3 \pm 8i$

31 Solve $x^2 + 8x + 65 = 0$.

a. $4 \pm 7i$

b. $-4 \pm 7i$

c. $-3 \pm 7i$

d. $-4 \pm 8i$

32 Solve $x^2 - 10x + 41 = 0$.

a. $5 \pm 4i$

b. $-5 \pm 4i$

c. $5 \pm 5i$

d. $6 \pm 4i$

33 Solve $x^2 - 4x + 20 = 0$.

a. $3 \pm 4i$

b. $-2 \pm 4i$

c. $2 \pm 5i$

d. $2 \pm 4i$

34 Solve $x^2 - 14x + 98 = 0$.

a. $8 \pm 7i$

b. $7 \pm 8i$

c. $-7 \pm 7i$

d. $7 \pm 7i$

35 Solve $x^2 - 6x + 45 = 0$.

a. $3 \pm 7i$

b. $4 \pm 6i$

c. $-3 \pm 6i$

d. $3 \pm 6i$

36 Solve $x^2 - 12x + 85 = 0$.

a. $6 \pm 7i$

b. $-6 \pm 7i$

c. $7 \pm 7i$

d. $6 \pm 8i$

37 Solve $x^2 - 16x + 113 = 0$.

a. $9 \pm 7i$

b. $8 \pm 8i$

c. $-8 \pm 7i$

d. $8 \pm 7i$

38 Solve $x^2 - 14x + 74 = 0$.

a. $7 \pm 6i$

b. $7 \pm 5i$

c. $8 \pm 5i$

d. $-7 \pm 5i$

39 Solve $x^2 + 16x + 80 = 0$.

- a. $8 \pm 4i$
 - b. $-7 \pm 4i$
 - c. $-8 \pm 4i$
 - d. $-8 \pm 5i$
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Multiple choice — hard

40 Find $|z|$ for $z = -7 + 24i$.

- a. 23
- b. 25
- c. 31
- d. 27

41 Find $|z|$ for $z = -1 + 6i$.

- a. 37
- b. $\sqrt{39}$
- c. $\sqrt{35}$
- d. $\sqrt{37}$

42 Find $|z|$ for $z = -1 - 4i$.

- a. $\sqrt{17}$
- b. $\sqrt{15}$
- c. 17
- d. $\sqrt{19}$

43 Find $|z|$ for $z = -5 + 12i$.

a. 15

b. 11

c. 13

d. 17

44 Find $|z|$ for $z = -12 - 35i$.

a. 47

b. 39

c. 35

d. 37

45 Find $(2 - 5i) - (4 - 2i) + (1 - i)$.

a. $1 - 4i$

b. $-1 - 6i$

c. $2 - 4i$

d. $-1 - 4i$

46 Find $(2 - 7i) - (-6 - 9i) + (-8 - 5i)$.

a. $0 - 5i$

b. $3 - 3i$

c. $0 - 3i$

d. $2 - 3i$

47 Find $(-7 - 4i) - (-8 - 9i) + (-7 + 9i)$.

a. $-4 + 14i$

b. $-6 + 14i$

c. $-6 + 12i$

d. $6 + 14i$

48 Find $(8 + i) - (5 - 3i) + (-8 + 9i)$.

a. $-5 + 11i$

b. $-5 + 13i$

c. $-3 + 13i$

d. $5 + 13i$

49 Find $(-2 + 5i) - (6 + 6i) + (-6 + 6i)$.

a. $-14 + 5i$

b. $-14 + 3i$

c. $14 + 5i$

d. $-12 + 5i$

50 Multiply $(8 + 4i)(1 + 3i)$.

a. $-4 + 28i$

b. $20 + 28i$

c. $-4 + 30i$

d. $-4 + 20i$

51 Multiply $(3 - i)(6 + 7i)$.

a. $25 + 27i$

b. $25 + 15i$

c. $25 + 17i$

d. $11 + 15i$

52 Multiply $(-5 + 2i)(-2 + i)$.

a. $8 - 7i$

b. $12 - 9i$

c. $8 - i$

d. $8 - 9i$

53 Multiply $(6 - 4i)(-9 + i)$.

a. $-50 + 44i$

b. $-50 - 30i$

c. $-50 + 42i$

d. $-58 + 42i$

54 Multiply $(9 - 8i)(8 - 6i)$.

a. $120 - 118i$

b. $24 - 118i$

c. $24 - 116i$

d. $24 + 10i$